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REVIEW ARTICLE

Validation status of cognitive digital assessments by the FDA BEST framework and context of use in preclinical AD studies: A systematic review

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Abstract

Digital cognitive assessments have rapidly expanded in Alzheimer's disease (AD) research, offering a sensitive, scalable, and cost-effective alternative to traditional neuropsychological tests. This systematic review examines the validation and utility of digital cognitive assessments in cognitively normal (CN) individuals and explores their potential classification within the US Food and Drug Administration's Biomarkers, Endpoints, and other Tools (FDA BEST) framework. Additionally, we provide recommendations to consider for their implementation. Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, we searched PubMed for studies validating digital cognitive tools against paper-based tests and standard AD biomarkers, including measures of amyloid beta and tau in fluid and neuroimaging biomarkers. Our findings suggest potential use as risk or monitoring biomarkers, though further longitudinal validation is needed. This review highlights the latest advancements in digital cognitive assessments, their role as novel AD biomarkers, and essential considerations for their effective use in AD.

KEYWORDS
Alzheimer's disease, amyloid beta, biomarkers, digital biomarkers, digital cognitive assessments, preclinical Alzheimer's disease

Highlights

- Digital cognitive assessments for preclinical Alzheimer's disease (AD) are associated with established biomarkers, including paper-based neuropsychological tests and amyloid beta and tau measures in both fluid and neuroimaging techniques.
- These assessments have the potential to serve as novel AD biomarkers classified within the US Food and Drug Administration's Biomarkers, Endpoints, and other Tools framework and context of use, but long-term studies spanning different disease stages are needed to fully establish their validity for some of the biomarker categories.
- Several biases may be present when conducting digital cognitive assessments; their optimal use should follow specific recommendations to minimize them.

1 | BACKGROUND

Alzheimer's disease (AD) is the primary cause of dementia, impacting ≈ 50 million individuals worldwide. It is characterized by distinctive neuropathological features, specifically the buildup of amyloid beta (Aβ) plaques and tau neurofibrillary tangles in the brain, which can develop up to 20 years prior to the onset of cognitive symptoms.[1] The latest revised criteria for diagnosis and staging of AD, released in 2024 by the Alzheimer's Association (AA), redefined AD as a spectrum that ranges from asymptomatic brain pathology—termed the preclinical stage—to the emergence and progression of cognitive and functional decline.[2] This underscores the importance of distinguishing between clinically identified phenotypes (clinical AD) and the underlying biological factors (biological AD). The AA also suggests that abnormalities in specific Core 1 biomarkers are sufficient for diagnosing AD. These biomarkers include positive Aβ positron emission tomography (PET) scans, cerebrospinal fluid (CSF) Aβ42/40 ratios, CSF phosphorylated tau (p-tau)181/Aβ42, CSF total tau (t-tau)/Aβ42, and "accurate" plasma assays defined as having performance levels equivalent to approved CSF assays in detecting abnormal Aβ PET in the relevant population.

During the preclinical stage, subtle cognitive decline may appear despite largely intact cognitive abilities. Detecting these early changes is crucial for clinical management and intervention (patient selection for clinical trials, monitoring purposes, etc.) because cognitively normal (CN) individuals with higher AD pathology tend to have more subtle cognitive deficits and experience greater cognitive decline, placing them at increased risk of progressing to the clinical stages of the disease.[3,4] However, traditional cognitive assessments, based on paper-and-pencil tests, present some limitations. They require the need for specialized personnel and in-person visits to the clinical site, lengthy administration times, high costs, and restricted measurements to discrete time points, making them less ecologically valid and potentially contributing to variability in results.

In response to these limitations, there has been growing interest in digital cognitive assessments to serve as novel biomarkers that can non-invasively and scalably monitor early-stage cognitive decline associated with AD. These tools fall under the broader category of digital biomarkers. However, as the definition of a digital biomarker is still evolving, there is currently no universal consensus, and recent reviews have shown significant variability in how the term is defined across studies.[5] In this paper, we adopt the definition proposed by the US Food and Drug Administration (FDA), which describes a digital biomarker as "a characteristic or set of characteristics, collected from digital health technologies, that is measured as an indicator of normal biological processes, pathogenic processes, or responses to an exposure or intervention, including therapeutic interventions."[6] Within this framework, digital cognitive assessments may be considered digital biomarkers. Although they do not directly measure pathology, they can serve as indicators of underlying pathogenic cognitive processes or responses to therapeutic interventions. However, to specifically qualify as biomarkers, such tools must demonstrate a strong and validated association with already established AD-related biological markers. These tools can be used both in a clinic and in home-based environments (without any additional personnel), are less costly, allow for more frequent and ecologically valid testing, and are capable of detecting subtle cognitive changes, thus improving diagnostic and follow-up accuracy in neurodegenerative diseases, including AD.[7–9] Within their realm, active and passive digital biomarkers represent two distinct approaches for capturing data related to cognitive function and other physiological measures. Active digital biomarkers are generated from data collected through direct interaction with digital devices or applications. These biomarkers require users to perform specific tasks designed to assess cognition, such as cognitive batteries, individual cognitive tests, or speech-based tasks, can be administered through various platforms—including smartphones, tablets, and computers, and can be deployed in both clinical and remote settings and with or without supervision.[8] In contrast, passive digital biomarkers are collected unobtrusively, without an intentional user interaction, as participants engage in daily activities. They are obtained through wearable devices, smartphones, or home-based sensors that monitor physiological and behavioral patterns or environmental sensors.[8,10] This review focuses specifically on active digital biomarkers (remote and in-clinic digital cognitive assessments), as our objective is to address cognitive performance and cognitive change.

RESEARCH-IN-CONTEXT

1. **Systematic review:** We conducted a comprehensive literature review using conventional search engines to examine studies using digital cognitive assessments in cognitively normal populations and with established Alzheimer's disease (AD) biomarker validation.
2. **Interpretation:** We included different digital tools used to enable both on-site and remote cognitive assessment in preclinical AD. Their validation against established AD biomarkers was found ranging from predominantly weak to moderate. Their use as biomarkers in research and clinical contexts is promising, especially as risk biomarkers. Potential future uses in different biomarker contexts have been identified and discussed. Recommendations for their optimal use were also provided. These findings have great implications for enhancing the optimal use of digital cognitive assessments.
3. **Future directions:** Future research should prioritize the longitudinal implementation of digital cognitive assessments to track disease trajectories and assess their association with AD pathology accumulation over time to validate their biomarker use in different categories.

Systematic reviews conducted in 2021[9] and 2024[11] have explored the role of digital cognitive assessments in preclinical AD, emphasizing the need for validation against established AD biomarkers. The present review summarizes recent developments in digital cognitive assessments focusing on their validation with gold-standard tests and established biomarkers: CSF, blood, PET, and magnetic resonance imaging (MRI). We also address the classification of these biomarkers according to their context of use (COU) as defined by the FDA BEST (Biomarkers, Endpoints, and other Tools) framework, to identify their most appropriate applications, and also propose recommendations for their use, while considering key factors, such as administration times, the setting in which they are applied (in clinic vs. remote), the required devices or platforms, whether they assess individual cognitive tasks or provide global cognitive insights, technology familiarity, language, and socioeconomic and demographic factors.

By integrating these considerations, we aim to provide a nuanced perspective on the use of these assessments as biomarkers according to their optimal uses.

To achieve these objectives, we first provide a brief summary of the literature on the associations between digital cognitive assessments and paper-and-pencil tests, as well as with AD pathology biomarkers, including CSF, plasma, PET, and MRI, presented through a summary table. Next, we examine the use of digital cognitive assessments within the framework of the FDA BEST classification. Finally, we offer the recommendations for their use.

2 | METHODS

We followed the new Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines,[12] to perform the following search and screening procedures (Figure 1).

2.1 | Search and identification strategies

We searched for original studies assessing the potential use of digital cognitive assessments in AD, especially in the preclinical phase of this disease, from April 2024 to January 2025. We searched the electronic database PubMed for relevant publications using the following search terms: "digital biomarkers" AND "preclinical AD" AND "Alzheimer's," "digital cognitive assessments" AND "preclinical AD" AND "Alzheimer's," "smartphone" AND "preclinical AD" AND "Alzheimer's," "tablet" AND "preclinical AD" AND "Alzheimer's."

The initial search yielded 1593 articles. All records were uploaded to Rayyan, a web-based tool for conducting systematic reviews.[13] Duplicate records were automatically removed by the software and further manually verified, resulting in 543 duplicates being excluded.

2.2 | Screening procedures

Studies were excluded if they were focused on other diseases and not specifically on AD populations only; among those that did focus on AD we excluded studies that examined only populations with mild cognitive impairment (MCI) or dementia, not having AD established biomarker validation, did not use active digital cognitive assessments, or relied on passive digital biomarkers. Additionally, we excluded animal studies, ongoing or incomplete studies, case reports, case series, modeling studies, review articles, systematic reviews, editorials, comments, letters, book chapters, conference abstracts, and opinion pieces. Only articles published in English were included.

After removing duplicates, 1053 articles remained for screening based on title and abstract using Rayyan. Of these, 961 were excluded for meeting one or more of the exclusion criteria described above.

Due to the novelty of this research area, and particularly in the context of preclinical AD populations, and the restrictive eligibility criteria, only 92 articles were screened full text for eligibility. From those, we excluded 64 for meeting one or more of the exclusion criteria described above (see Figure 1 for details), resulting in 27 studies being included in the final review. Two reviewers independently assessed the full-text articles for eligibility. Disagreements were resolved through discussion or by consulting a third reviewer.

Due to the heterogeneity and methodological differences among the included studies, we could not conduct a meta-analysis and instead we performed a narrative synthesis.

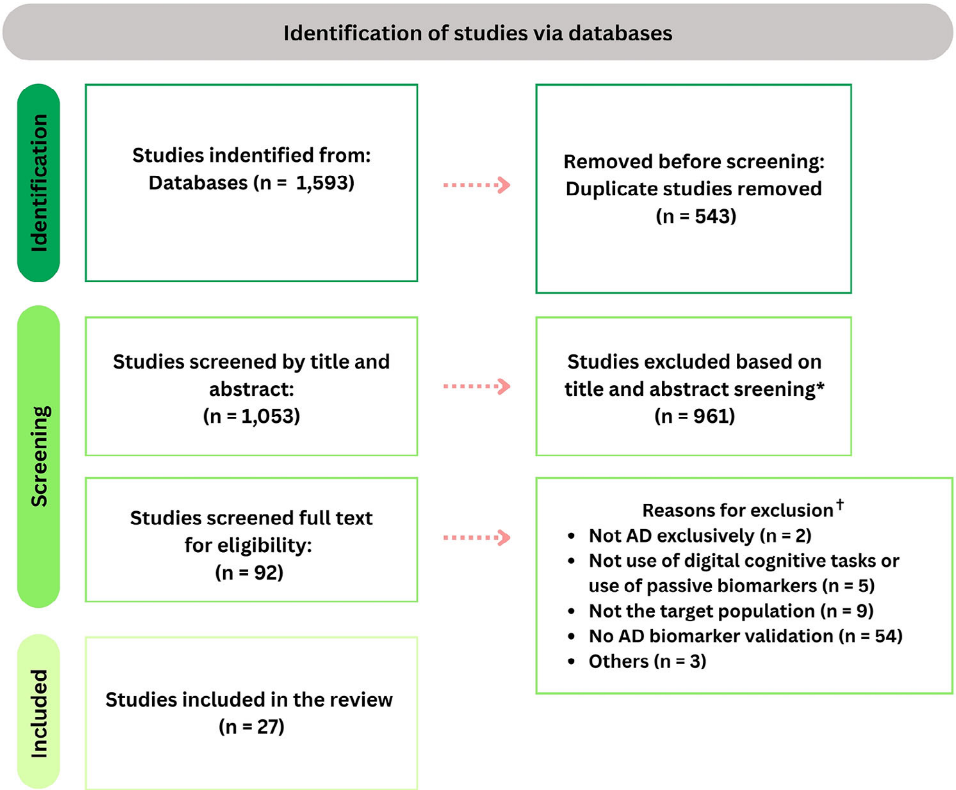

FIGURE 1 PRISMA 2020 guidelines flow diagram. Note. *Exclusion based on the exclusion criteria explained in section 2.2. †Some articles were excluded for more than one reason; therefore, the number of articles per exclusion reason differs from the total number of excluded articles. AD, Alzheimer's disease; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

2.3 | FDA BEST biomarker classification

This review aims to go beyond the validation of digital cognitive assessments through neuropsychological and AD biomarkers by exploring their broader potential in both clinical practice and research. With the rapid rise in technology and digital device usage, integrating these tools into clinical trials offers promising applications and advantages.

To provide a comprehensive framework for their use in this context, we propose a classification system based on a joint initiative led by FDA and the National Institutes of Health (NIH) named BEST biomarker classification. The BEST system defines seven key biomarker categories—susceptibility/risk, diagnostic, monitoring, prognostic, predictive, pharmacodynamic/response, and safety—each associated with a specific COU[14] (Table 1). The COU is a concise description of a biomarker's intended role in drug development, comprising two elements: (1) the relevant BEST biomarker category and (2) the biomarker's specific purpose in drug development. Some biomarkers may serve multiple COUs; for example, Aβ PET can function as a diagnostic, monitoring, or a response biomarker depending on the study nature and objectives. Thus, it is important to recognize that these classifications are not rigid but can be adaptable and flexible depending on the specific goals and endpoints of each study. Digital cognitive assessments, still evolving, can similarly adapt to various study aims, designs, and populations, often overlapping depending on the study needs.[15,16]

2.4 | Previous considerations for the integration of digital cognitive assessments within the FDA BEST framework

Before categorizing digital cognitive assessments within the BEST framework, it is essential to define the criteria used to characterize AD, given the existence of different approaches. This review adopts the newly revised criteria published,[2] which define AD biologically rather than based on syndromic presentation. The criteria focus on core AD biomarkers, which are those in the A (Aβ) and T (tau) categories, and are measured through CSF, plasma, or PET (Table 2). According to these updated guidelines, an abnormal Core 1 biomarker result is considered sufficient to establish a diagnosis of AD, and includes specifically abnormality in the A, T1, or hybrid ratio biomarkers of Table 2. Unimpaired individuals with abnormal biomarker findings are therefore considered to have the disease and at risk for developing AD-related symptoms, rather than at risk for the disease itself. However, for clinical purposes the diagnosis should remain for only symptomatic individuals at this time.

This biologically driven approach aligns with the increasing focus on disease-modifying therapies, many of which are under evaluation in clinical trials targeting the early stages of AD. However, alternative frameworks remain equally valid, and the choice of criteria can influence biomarker classification within the BEST model. Therefore,

TABLE 1 Biomarker classification and COUs based on the FDA BEST classification framework.

Biomarker	Measurement	Context of use	AA definition
Susceptibility/ risk	Indicates the potential for developing a disease or medical condition in an individual who does not currently have clinically apparent disease or the medical condition	• Indicate the potential for developing a disease or sensitivity to an exposure	Abnormality in Core 1 AD biomarkers. Individuals are at risk for developing cognitive symptoms.
Diagnostic	Detect or confirm the presence of a disease or condition of interest or identify individuals with a subtype of the disease	• Patient selection	Abnormality in Core 1 AD biomarkers for biological diagnosis. For clinical use, cognitive symptoms are needed.
Monitoring	A biomarker measured repeatedly for assessing status of a disease or medical condition or for evidence of exposure to (or effect of) a medical product or an environmental agent	• Detect a change in the degree or extend of a disease • Indicate toxicity or assess safety • Provide evidence of exposure	–
Prognostic	Identify the likelihood of a clinical event, disease recurrence, or progression in patients who have the disease or a medical condition	• Stratify patients • Enrichment: inclusion/exclusion data	–
Predictive	Identify individuals who are more likely than similar individuals without the biomarker to experience a favorable or unfavorable effect from exposure to a medical product or an environmental agent	• Identify individuals on the basis of effect from a specific intervention or exposure	–
Response	Show that a biological response, potentially beneficial or harmful, has occurred in an individual who has been exposed to a medical product or an environmental agent	• Efficacy biomarker/surrogate endpoint • Show biological response related to an intervention or exposure	–
Safety	A biomarker measured before or after an exposure to a medical product or an environmental agent to indicate the likelihood, presence, or extent of toxicity as an adverse effect	• Indicate the presence or extent of toxicity related to an intervention or exposure	–

Abbreviations: AA, Alzheimer's Association; AD, Alzheimer's disease; BEST, Biomarkers, Endpoints, and other Tools; COU, context of use; FDA, US Food and Drug Administration.

TABLE 2 Core 1 AD biomarkers for diagnosis based on revised criteria by Jack et al.[2]

Intended use	CSF	Plasma	Imaging
Diagnosis			
A: (Aβ proteinopathy)	–	–	Amyloid PET
T1: (phosphorylated and secreted AD tau)		p-tau217	–
Hybrid ratios	p-tau181/Aβ42, t-tau/Aβ42, Aβ42/40	%p-tau217	–

Abbreviations: Aβ, amyloid beta; AD, Alzheimer's disease; CSF, cerebrospinal fluid; PET, positron emission tomography; p-tau, phosphorylated tau; t-tau, total tau.

it should be carefully considered based on the study's objectives and research perspective.

3 | RESULTS

3.1 | Neuropsychological and AD biomarker validation

The following results (Table 3) are based on a review of 27 articles that used digital cognitive assessments in CN populations and validated them against established AD biomarkers including: Altoida;[17] The Ambulatory Research in Cognition (ARC);[18] Boston Remote Assessment for Neurocognitive Health (BRANCH);[19] BRANCH Multi-Day;[20] Paired Associate Learning (PAL) task from the Cambridge Neuropsychological Test Automated Battery (CANTAB);[21] Cognitron;[22] Computerized Cognitive Composite (C3);[23,24] Cogstate Brief Battery (CBB);[25] Computerized PAL (CPAL)[26] from Cogstate; Connected Speech and Language (CSL);[27] the DCTclock;[28,29] Face Name Associative Memory Exam (FNAME) from the NIH Toolbox Cognition Battery (NIHTB-CB);[30] FNAME from C3;[31] Mobile Monitoring of Cognitive Change (M2C2);[32] the Mayo Test Drive (MTD) platform;[33]

TABLE 3 Summary of the results from the digital cognitive assessments included in this review and their possible biomarker category.

Digital Assessment	Device	Administration setting	Study design	Population (n)	AD biomarker	Main outcomes	FDA BEST possible biomarker category	Reference
Altoida	Tablet	Remote and in clinic 10–15 min ≈	Cross-sectional and longitudinal (One time point for in-clinic, 3 time points for remote)	CN (84) Prodromal AD (37) 56 completed remote testing (42 CN)*	Aβ CSF Aβ PET	Digital in-clinic score correlated with the paper-and-pencil score (r = 0.56). Digital at-home score distinguished Aβ+/− in CN (AUC = 0.76). Lower scores in Aβ+ than Aβ− (β = −0.40).	Risk Monitoring* Diagnostic*	(Muurling et al., 2023)[17]
ARC	Smartphone	Remote 3/4 times per day - 7 days < 3 min ≈	Cross-sectional and longitudinal (a subset completed follow-ups in 6 months and 1 year)	CN (268) Mild dementia (22)	CSF (Aβ40, Aβ42, t-tau, p-tau181) Aβ and tau PET (FBP; PiB; FTP) MRI	ARC composite correlated with a standard cognitive composite (r = −0.47 for CN). Worse performance correlated with CSF Aβ42 (r = −0.23), PET Aβ (r = 0.26), tau PET (r = 0.11), CSF t-tau (r = 0.28), CSF p-tau181 (r = 0.25) and CSF p-tau181/Aβ42 (r = 0.29), in a similar manner as the standard global composite.	Risk Monitoring*	(Nicosia et al., 2023)[18]
BRANCH	Any device (primarily smartphone)	Remote 22 minutes ≈	Cross-sectional (retest for a subset)	CN (234)	Aβ and tau PET (PiB; FTP)	Correlated with PACC (r = 0.62), along with higher cortical Aβ (r = −0.205), and higher entorhinal tau (r = −0.178).	Risk Monitoring*	(Papp et al., 2021)[19]
BRANCH - Multi-day	Any device (primarily smartphone)	Remote 12 minutes/day ≈ 7 days	Longitudinal	CN (164)	Aβ PET (PiB)	Correlates with PACC (r = 0.54). Aβ+ showed diminished learning curves at day 4 (d = 0.49). Diminished learning curves were associated with greater annual PACC5 decline (r = 0.54).	Risk Monitoring* Prognostic*	(Papp et al., 2024)[20]

(Continues)

TABLE 3 (Continued)

Digital Assessment	Device	Administration setting	Study design	Population (n)	AD biomarker	Main outcomes	FDA BEST possible biomarker category	Reference
CANTAB - PAL	Tablet	In clinic < 10 minutes ≈	Cross-sectional	CN (62) MCI (11)	Aβ PET (PiB) MRI	Higher CPAL error scores were associated with Aβ positivity, with moderate magnitude (effect size = 0.60).	Risk	(Pettigrew et al., 2022) [21]
Cognitron	Computer	Remote 30/40 minutes ≈	Cross-sectional	CN (255)	Aβ PET (FBP) MRI	Cognitron composites correlated with paper-and-pencil tests (r = 0.50). Poorer performance predicted Aβ positivity (OR = 1.79) and WMHV (OR = 1.23). Brain and hippocampal atrophy rates correlated with poorer visuospatial performance (β = −0.42) and accuracy (β = −0.01).	Risk Diagnostic*	(Giovane et al., 2025) [22]
Cogstate - C3	Tablet	In clinic Supervised 15–20 minutes ≈	Longitudinal (2 visits)	CN (4486)	Aβ PET (FBP)	Performance correlated with PACC (r = 0.39). Aβ+ individuals performed worse on the C3 compared to Aβ in both visits (d = −0.22).	Risk Monitoring* Response*	(Papp et al., 2021) [23]
Cogstate - C3	Tablet	Remote 15–20 minutes ≈	Longitudinal (monthly for 1 year)	CN (114)	Aβ and tau PET (PIB; FTP)	3-month C3 slopes correlated with annual change on the PACC5 (r = 0.69). Lower practice effects correlated with higher global Aβ burden (r = −.20), entorhinal and inferior temporal tau PET burden (r = −0.38; −0.23). 3-month C3 slopes identified PACC5 decliners (AUC = 0.91).	Risk Monitoring Diagnostic* Prognostic* Response*	(Jutten et al., 2022) [24]

(Continues)

TABLE 3 (Continued)

Digital Assessment	Device	Administration setting	Study design	Population (n)	AD biomarker	Main outcomes	FDA BEST possible biomarker category	Reference
Cogstate - CBB	Computer	In clinic 10 minutes ≈	Cross-sectional	CN (338) MCI (185) AD (7)	Aβ and tau PET (FBP; FBB; FTP)	It differentiated CN from AD and from MCI (AUC = 0.87; 0.70), with one individual task achieving an AUC = 0.92 to differentiate CN from AD. Predicted conversion from CN to MCI over 2 years (AUC = 0.74). Performance was associated with longitudinal cognitive decline ($r_p = 0.25$), and predicted, in Aβ+ CN +, future entorhinal tau ($r_p = -0.26$).	Risk Prognostic Diagnostic*	(Vanderlip et al., 2024) [25]
Cogstate - CPAL	Computer / tablet	In clinic < 10 minutes ≈	Longitudinal (3 assessments over 36 months: baseline, and 18- and 36-month follow-ups)	CN (356)	Aβ PET	Aβ– task performance improved over time, whereas Aβ+ showed no practice effect (d = 0.30).	Risk Monitoring	(Baker et al., 2019) [26]
CSL	Audio recording	In clinic 2 minutes ≈	Longitudinal (biennial)	CN (247) MCI (6) AD (2)	Aβ PET (PiB) MRI	Aβ positivity was associated with a more rapid decline in the use of specific content words biennially (β = --0.26).	Risk Prognostic	(Mueller et al., 2021) [27]
DCTclock	Digital pen	In clinic < 10 minutes ≈	Cross-sectional	CN (264) MCI (36)	Aβ and tau PET (PiB; FTP)	It had an excellent discrimination between diagnostic groups (AUC = 0.86). In CN, it discriminated between A$\beta\pm$ groups whereas hand-scored did not and with better discrimination (AUC = 0.72) than the PACC (AUC = 0.63), poorer performance was associated with higher Aβ burden ($r = -0.24$) and higher levels of entorhinal tau ($r = -0.19$). It correlated with regions with Aβ in frontoparietal and default mode networks (βs=| −0.35|–|-0.10|).	Risk Diagnostic*	(Rentz et al., 2021) [28]

(Continues)

TABLE 3 (Continued)

Digital Assessment	Device	Administration setting	Study design	Population (n)	AD biomarker	Main outcomes	FDA BEST possible biomarker category	Reference
DCTclock	Digital pen	In clinic < 10 minutes ≈	Cross-sectional	CN (1633) MCI (23)	MRI	All dCDT composite scores were associated with total cerebral brain volume (βs = |0.099|-|−0.080|), most with cerebral white matter volume (βs=|0.084|-|−0.069|) and one with gray matter volume (β=0.059). dCDT scores, MRI measures, and clinical risk factors had an AUC of 0.897 to distinguish MCI from CN.	Diagnostic*	(Yuan et al., 2022) [29]
FNAME - NIHTB-CB	Tablet	In clinic	Cross-sectional	CN (103) MCI (17) AD (2)	CSF (Aβ42/ p-tau181 ratio) Aβ PET (FBP; PiB)	In CN only, no associations or differences were found, suggesting FNAME does not reliably distinguish Aβ status in this group.	For CN individuals it would not classify for any based on this study.	(Rentz et al., 2023) [30]
FNAME - C3	Tablet	Remote 20 min ≈	Longitudinal (monthly during 1 year)	CN (94)	Aβ and TAU PET (PiB; FTP)	Reduced learning curves with repeated exposures correlated with tau burden (r = −0.22), Aβ+/Aβ− differences appeared by the fourth exposure (d = 0.66).	Risk Monitoring* Prognostic*	(Samaroo et al., 2020) [31]
M2C2 Tab-CAT DCTclock	Smartphone Tablet - In clinic Digital pen - In clinic	M2C2: Remote - 3/4 minutes ≈	Cross-sectional and longitudinal (M2C2 8 days; Tab-CAT and DCTclock one time point)	CN (69)	Aβ PET	M2C2 tasks, TabCAT, and DCTclock correlated with MoCA (r = −0.37 to 0.46). The Prices task best distinguished Aβ status (AUC = 0.77), while TabCAT had the highest in-clinic AUC (0.76), followed by MoCA (0.74) and DCTclock (0.73).	Risk Diagnostic* Monitoring*	(Thompson et al., 2023) [32]

(Continues)

TABLE 3 (Continued)

Digital Assessment	Device	Administration setting	Study design	Population (n)	AD biomarker	Main outcomes	FDA BEST possible biomarker category	Reference
MTD	Computer/ multi-device	Remote	Cross-sectional	CN (643) MCI/dementia (41)	Aβ and tau PET (PiB; FTP) MRI	Composites and individual tasks correlated with paper-and-pencil tests of the same domain (r's = |-0.63|–|0.68|). In CN, all neuroimaging biomarkers (Aβ meta-ROI, tau meta-ROI, tau EC-ROI, HV, and WMH volume) correlated with MTD measures (r's = |-0.37|–|0.36|). MCI/dementia participants had worse performances compared to CN with large effect sizes (> 0.8)	Risk	(Boots et al., 2024) [33]
neotiv (MDT-O, ORR-DR, MDT-S)	Smartphone	Remote 20 minutes + 30 minutes delay ≈	Longitudinal (biweekly assessments for 12 months)	CN (76) MCI (23)	CSF (Aβ42; Aβ40) Plasma (p-tau217) Aβ and tau PET (FMM; FTP) MRI	Correlated with PACC (r = 0.63; 0.70). In combination with plasma p-tau217 improved mPACC decline prediction across 5 years (AIC = −940.50). Performance associated with tau PET in area 35 (βs = −0.71; −0.32; −0.06) and in anterior hippocampus (βs = −0.44; −0.017; −0.02), and with cortical thickness (βs = 0.22; 0.23). MDT-S correlated with anterior/posterior hippocampal volume (βs = 282.4; 215.53), while ORR-DR to posterior (β = 14.8). Associated with Aβ PET (βs = −0.29; −0.12; −0.02).	Risk Prognostic Diagnostic* Monitoring*	(Berron et al., 2024) [34]

(Continues)

TABLE 3 (Continued)

Digital Assessment	Device	Administration setting	Study design	Population (n)	AD biomarker	Main outcomes	FDA BEST possible biomarker category	Reference
NIHTB-CB	Tablet	In clinic 30 minutes ≈	Cross-sectional	CN (118)	Aβ and tau PET (PiB; FTP) MRI	No NIHTB-CB measures were associated with Aβ deposition. Some NIHTB-CB measures were associated with tau deposition in higher Braak regions (βs = |-0.19|–|-0.28|).	Risk	(Snitz et al., 2020)[35]
Novoic	Audio recording on personal smartphone	Remote < 20 minutes ≈	Longitudinal (7–8 days)	CN (66) MCI (49)	CSF (Aβ) Aβ PET	Cognitive performance correlated with PACC ($r = 0.74$); detected Aβ positivity in the CN subsample (AUC = 0.74).	Risk Diagnostic* Monitoring*	(Fristed et al., 2022)[36]
OCTAL	Computer/tablet	Remote	Cross-sectional	CN (46) AD (53)	Plasma (p-tau181)	Correlates with plasma p-tau181 ($r = 0.34$). For group classification, the best model combined OCTAL variables (AUC = 1) and to predict p-tau181 levels, the most effective model integrated Aβ42/40 ratio with OCTAL scores ($R^2 = 0.50$).	Risk Diagnostic*	(Toniolo et al., 2024)[37]
ORCA-LLT	Any device	Remote 20 minutes ≈	Longitudinal (6 days)	CN (80)	Aβ PET MRI	Performance predicted baseline memory scores ($\beta = 0.26$) and past memory scores ($\beta = 0.27$). In Aβ+ CN learning curves were diminished compared to Aβ– (d = 2.22), predicted Aβ PET ($\beta = -0.23$) and were associated with smaller hippocampal and larger ventricular volumes.	Risk Monitoring* Diagnostic*	(Lim et al.,2020)[38]

(Continues)

TABLE 3 (Continued)

Digital Assessment	Device	Administration setting	Study design	Population (n)	AD biomarker	Main outcomes	FDA BEST possible biomarker category	Reference
SLS (MTD platform)	Computer/multi-device	Remote 15 minutes ≈	Cross-sectional	CN (326) MCI (21) Dementia (5)	Aβ and tau PET (PiB; FTP)	Correlates with paper-and-pencil tests of the same domain (r = 0.62). In CN, SLS predicted Aβ positivity (AUC = 0.63), and A+T+ status (AUC = 0.69), with adjustments for demographics enhancing accuracy (AUC = 0.77 and 0.83).	Risk Diagnostic*	(Stricker et al., 2024) [39]
SMT and 4MT tasks	Tablet	Semi-automated In clinic 5–15 minutes ≈	Cross-sectional	CN (1875)	CSF (Aβ42, p-tau181) MRI	Both correlated with a paper-and-pencil test (r = 0.25; 0.30). Poorer scores were linked to higher levels of p-tau181 (βs = 0.03; –0.04), lower Aβ42 (β = 0.03), lower p-tau181/Aβ42 ratio (βs = –0.04; –0.05). Greater performance was associated with larger volumes on the EC (βs = 0.06; 0.03), hippocampus (β = 0.03) and Pcun (β = 0.03).	Risk	(Coughlan et al., 2023)[40]
Speech and ML algorithms	Audio recording on iPod	In clinic 1–2 minutes ≈	Cross-sectional	CN (92) MCI (114)	CSF (Aβ42, t-tau, p-tau181) MRI	Baseline cognitive performance was associated with change in CDR-SOB after 2 years (βs = –0.18; 0.26). Lexical-semantic scores differed between Aβ+/Aβ– in the whole sample, detected Aβ+ status in the whole sample (AUC = 0.77) and in CN only (AUC = 0.61). Acoustic scores were associated with hippocampal volume (R^2 = 0.03).	Risk Diagnostic* Predictive* Prognostic*	(Hajjar et al., 2023)[41]

(Continues)

TABLE 3 (Continued)

Digital Assessment	Device	Administration setting	Study design	Population (n)	AD biomarker	Main outcomes	FDA BEST possible biomarker category	Reference
Spontaneous Speech - Open Ended Tasks	Audio recording during tasks	In clinic 5 minutes ≈	Cross-sectional	SCD (63)	CSF (Aβ42) Aβ PET (FBB, FBP)	Linguistic parameters correlated with paper-and-pencil tests of the same domain ($r = 0.34; 0.25; -0.29$). High Aβ burden was associated with some speech parameters as fewer concrete nouns (OR = 7.6), content words ($\beta = -0.54$), increased use of abstract nouns ($\beta = 0.69$), and increased D-level ($\beta = 0.48$).	Risk	(Verfaillie et al., 2019) [42]
TabCAT-BHA	Tablet - supervised	In clinic Supervised 10–15 minutes ≈	Cross-sectional	SCD (49) MCI (159) Dementia (101)	Plasma (p-tau181)	BHA subtests showed moderate to high correlations with standard tests of the same domain. Associated with plasma Aβ42/40 (β=0.196) and p-tau181 ($\beta = -0.271$). In SCD and MCI subsample it had an AUC = 0.883 to detect Aβ PET positivity. Plasma p-tau181, NfL, Aβ42/40, and TabCAT-BHA achieved the optimal accuracy (AUC = 0.962). Lower baseline TabCAT-BHA cognitive composite values were associated with greater increases in disease severity ($\beta = -0.389$).	Risk Prognostic Diagnostic*	(Tsoy et al., 2024)[43]

Abbreviations: 4MT, 4 Mountains Task; AD, Alzheimer's disease; AIC, Akaike information criterion; ARC, The Ambulatory Research in Cognition; AUC, area under the curve; Aβ, amyloid beta; BRANCH, Boston Remote Assessment for Neurocognitive Health; C3, Computerized Cognitive Composite; CANTAB, Cambridge Neuropsychological Test Automated Battery; CBB, Cogstate Brief Battery; CN, cognitively normal; CPAL, Computerized Paired Associate Learning task; CSF, cerebrospinal fluid; CSL, Connected Speech and Language; EC, entorhinal cortex; FBB, Florbetaben; FBP, Florbetapir; FDG-PET, Fluorodeoxyglucose Positron Emission Tomography; FNAME, Face Name Associative Memory Exam; FTM, Flutemetamol; FTP, Flortaucipir; HV, hippocampal volume; M2C2, Mobile Monitoring of Cognitive Change; MCI, mild cognitive impairment; MDT-O, Mnemonic Discrimination for Objects; MDT-S, Mnemonic Discrimination for Scenes; MoCA, Montreal Cognitive Assessment; MRI, magnetic resonance imaging; MTD, Mayo Test Drive; NfL, neurofilament light chain; NIHTB-CB, The National Institutes of Health Toolbox Cognition Battery; OCTAL, Oxford Cognitive Testing Platform; OR, odds ratio; ORCA-LLT, Online Repeatable Cognitive Assessment—Language Learning Test; ORR-DR, Object-in-Room - Delayed Recall; PACC, Preclinical Alzheimer's Cognitive Composite; PAL, Paired Associate Learning; Pcun, precuneus cortex; PET, positron emission tomography; PiB, Pittsburgh compound B; p-tau, phosphorylated tau; ROI, region of interest; SCD, subjective cognitive decline; SLS, Stricker Learning Span; SMT, SuperMarket Task; TabCAT-BHA, Tablet-based Cognitive Assessment Tool Brain Health Assessment; t-tau, total tau; WMH, white matter hyperintensity; WMHV, white matter hyperintensity volume.

*Not the Biomarker context of use validated in the study, but potential future use. **Diagnostic**: in combination with AD Core 1 Biomarkers. **Monitoring, Response, Prognostic, Predictive:** in need of longitudinal studies.

TABLE 4 FDA BEST biomarker category status for digital cognitive assessments.

Biomarker category (FDA BEST)	Status
Risk	Supported by studies
Monitoring	Supported by studies
Diagnostic	More studies needed
Prognostic	More studies needed
Response	More studies needed
Predictive	No studies yet
Safety	No studies yet

Green: Studies showed their use in that biomarker category.
Orange: Few studies showed their use in that biomarker category.
Red: No studies have shown their use in that biomarker category.
Abbreviations: BEST, Biomarkers, Endpoints, and other Tools; FDA, US Food and Drug Administration.

neotiv;[34] NIHTB-CB;[35] Novoic;[36] the Oxford Cognitive Testing Platform (OCTAL);[37] the Online Repeatable Cognitive Assessment–Language Learning Test (ORCA-LLT);[38] Stricker Learning Span (SLS) from the MTD platform;[39] SuperMarket Task (SMT) and 4 Mountains Task (4MT);[40] speech measures and machine learning (ML) algorithms;[41] spontaneous speech from open ended tasks;[42] and the Tablet-based Cognitive Assessment Tool Brain Health Assessment (TabCAT-BHA).[43] A potential biomarker classification is also provided based on the nature of the study and the application of the digital cognitive assessment within that context. Additionally, each tool's potential future use as a biomarker is indicated with an asterisk (*), signifying that while these tools have not been validated for that specific purpose in the study, they show promise for further validation in that capacity.

Table 4 provides the current status of each biomarker category based on the studies included.

3.2 | FDA BEST biomarker classification for digital cognitive assessments

3.2.1 | Diagnostic biomarkers

While digital cognitive assessments alone cannot provide a definitive diagnosis for AD, as abnormality in Core 1 biomarkers is required, none of the studies in this review validate them as sole diagnostic biomarkers. However, they demonstrate significant potential to complement diagnostic processes and enhance existing workflows.

The core AD biomarkers defined for biological diagnosis include amyloid PET, CSF Aβ42/40, CSF p-tau181/Aβ42, CSF t-tau/Aβ42, and plasma assays with accuracy comparable to approved CSF tests for detecting abnormal amyloid PET in the intended-use population (Table 2).[2] Integrating digital cognitive assessments—that are highly sensitive to early cognitive symptomatology—with these biomarkers holds significant promise for improving current diagnostic workflows in clinical routine as well as in research contexts and for refining patient selection. Digital tools with strong predictive performance, particularly those with high area under the curve (AUC), positive predictive value (PPV), and accuracy for biomarker positivity detection as well as those with sensitivity to cognitive deficits and high accuracy in distinguishing diagnostic groups (e.g., CN vs. MCI/AD), may contribute to more precise and cost-effective AD diagnosis.

Moreover, with plasma biomarkers now integrated into the revised AD criteria and building recognition for diagnostic use, their combination with digital cognitive assessments could even further streamline the diagnostic process, particularly improving accessibility, making the process able to be expanded to broader populations; cost effective; and non-invasive, as it may reduce reliance on more invasive techniques such as PET scans or CSF testing. Studies in this review highlight this latter potential—Berron et al.[34] found that neotiv tasks combined with plasma p-tau217 significantly improved 5 year prediction accuracy of cognitive decline in CN individuals. Similarly, the TabCat-BHA tool, when combined with plasma p-tau181, demographic data, and clinical variables, achieved the highest accuracy in detecting Aβ PET positivity.[43]

3.2.2 | Susceptibility/risk biomarkers

Given that this review focuses primarily on populations including CN individuals, most of the digital tools discussed herein can be classified as susceptibility/risk biomarkers under the revised criteria outlined earlier. This classification is justified by their strong associations with Core 1 AD biomarkers of Aβ and tau (Table 2), and therefore applies to digital tools that can reliably differentiate individuals with core AD biomarker positivity, as well as those that demonstrate continuous and significant associations with these biomarkers. These associations support the specific risk biomarker's COU of identifying an individual's risk of disease progression, particularly, in this case, the likelihood of developing cognitive symptoms due to AD in later stages.

Given this potential COU, digital cognitive assessments offer high utility in stratifying preclinical AD populations and identifying at-risk individuals for clinical trials, treatment interventions, and strategies targeting early disease stages, before the onset of clinical symptomatology. This is especially relevant as current disease-modifying treatments have demonstrated greater efficacy when administered in these preclinical phases, whereas later stage interventions have often shown diminished therapeutic effects in AD.[44]

While risk biomarkers are useful for identifying CN individuals at increased risk of developing cognitive symptoms, they do not predict the rate or severity of progression, as they only indicate a predisposition. To assess disease trajectory and progression, prognostic biomarkers are needed, requiring longitudinal studies to accurately track both neuropathological changes and cognitive decline over time.

3.2.3 | Prognostic biomarkers

To fully validate digital cognitive assessments as prognostic biomarkers, longitudinal studies are essential. While cross-sectional studies establish associations between cognitive performance and AD biomarkers (Aβ and tau), aiding in identifying at-risk individuals, longitudinal data allow for predicting disease trajectory and future cognitive outcomes, which is key for prognostic use.

While it is well known that individuals with AD pathology are at higher risk of AD-related progression, like neurodegeneration as well as progression to more advanced clinical stages,[45] validating prognostic biomarkers requires long-term observation to accurately map the disease trajectory. Digital tools that effectively detect Aβ and tau abnormalities or/and predict cognitive decline in subsequent years hold promise as prognostic biomarkers.

Currently, most studies included in this review rely on cross-sectional designs or short-term longitudinal data, limiting the available evidence for their prognostic utility. However, some studies have demonstrated that baseline performance on digital cognitive assessments is associated with future cognitive decline, as measured by standard neuropsychological tests such as the Preclinical Alzheimer's Cognitive Composite (PACC5) and the Clinical Dementia Rating Sum of Boxes (CDR-SOB).[20,24,25,34,41,43] For instance, neotiv reported that the combination of the digital assessment with plasma p-tau217 significantly contributed to the prediction of future decline,[34] while Cogstate CBB demonstrated predictive validity for conversion from CN to MCI within a 2 year timeframe, achieving an AUC of 0.74.[25] This may be effective in the COUs of this category: stratifying individuals based on their possible disease progression, aiding in the identification of those with a potentially poorer prognosis. Additionally, these biomarkers can enhance the enrichment of clinical trial cohorts based on expected prognosis when used as part of inclusion and exclusion criteria, because in clinical trials, prognostic biomarkers are normally used as entry criteria to identify patients who are most likely to progress during the trial.[16] However, as previously mentioned, longitudinal studies are crucial to fully establish the prognostic validity of digital cognitive assessments. Extended follow-up periods will provide essential insights into disease progression, linking cognitive performance to neuropathological changes and long-term clinical outcomes.

However, it is important to note that the conclusions drawn in these articles are based on a population with varying biomarker positivity, rather than being exclusively limited to those with abnormal Core 1 AD biomarkers (biologically diagnosed individuals). This distinction differs from the FDA's definition of prognostic biomarkers, which applies strictly to individuals already diagnosed with the disease. Nonetheless, we believe these findings still provide valuable prognostic insights into cognitive prediction and trajectories using digital cognitive assessments, offering meaningful information about potential disease progression even in preclinical stages.

3.2.4 | Monitoring biomarkers

Digital cognitive assessments are well suited for monitoring purposes compared to paper-and-pencil tests, offering advantages such as extensive data collection across multiple time points. They facilitate the specific COU of detection of subtle cognitive changes during disease progression, evaluation of treatment efficacy, and identification of cognitive patterns requiring intervention. Remote and unsupervised at-home assessments are particularly advantageous, as they reduce the need for frequent in-clinic visits, enhancing the ability to track progression during more time points. Several digital cognitive assessments reviewed in these studies show promise as monitoring biomarkers, especially those involving repeated measures. However, to be fully effective in a monitoring context, digital cognitive assessments should demonstrate test–retest reliability, high adherence, minimal practice effects, and associations with core AD biomarkers. Crucially, they must also show sensitivity to change, whether related to disease progression or the impact of an intervention.

The ARC composite, administered over 7 to 8 days, exhibited excellent between-person reliability (> 0.85) and test–retest reliability at both 6 month and 1 year follow-ups (intraclass correlations [ICCs] > 0.85), with high adherence rates and low dropout rates.[18] Similarly, the Multi-day BRANCH composite showed excellent test–retest reliability (ICC = 0.94) and high participant engagement.[20] The FNAME task within the C3 battery also had low discontinuation rates.[31] Other digital tools, such as ORCA-LLT, Cogstate CBB, and CPAL, have been administered at different time points and have been associated with AD core biomarkers;[25,26,38] however, further studies are needed to evaluate their adherence, test–retest reliability, or practice effects. Interestingly, in the Altoida study from 2023, which involved at-home testing, the test–retest reliability was initially poor (ICC = 0.48), but when stratified by phone type, the ICC for Android users was higher (0.70) compared to iOS users (0.33).[17] The neotiv assessment, which was conducted biweekly over a 12 month period, showed associations with AD pathology and accuracy in predicting cognitive decline over a 5 -year period (Akaike information criterion = −940.50), and also exhibited similar values regarding test–retest reliability (ICCs = 0.65–0.83), as well as high adherence rates.[34] Similarly, speech-based tools enhanced with artificial intelligence (AI) also showed promise as monitoring biomarkers, like the Novoic task administered over 7 days through a smartphone app exhibited similar results, enabling automated analysis without needing in-person interaction.[36]

Nevertheless, especially in contexts such as clinical trials, more longitudinal studies are needed. Most current studies cover only short time periods (from 7–8 days to 1 year), which limits the ability to draw conclusions about their utility in detecting meaningful cognitive changes. Therefore, longer term studies are essential to assess their sensitivity to change over time. In addition, further evaluations of test–retest reliability and adherence are required to ensure these tools can be reliably used across multiple time points for effective monitoring.

3.2.5 | Predictive biomarkers

None of the digital cognitive assessments reviewed have been fully validated as predictive biomarkers for selecting individuals for drug or treatment trials, because the identification of predictive biomarkers typically requires a comparison of the drug's effects on the outcome between individuals with and without the biomarker abnormality in randomized clinical trials, and no study in this case provided that. However, they do demonstrate significant potential in this context. Cognitive scores from these assessments have been linked to core AD biomarkers, with lower scores often correlating with higher pathology of Aβ and/or tau. This suggests that individuals with lower cognitive performance are the ones who may be more likely to benefit from anti-amyloid or anti-tau modifying drugs, as well as from preventive strategies and treatment. By identifying individuals with a higher likelihood of biomarker positivity and abnormality, particularly using digital tools with strong predictive capabilities, digital assessments can improve trial enrollment efficiency, reserving the need of costly and invasive biomarker techniques for more advanced steps in the process. This approach would minimize costs and logistics and potentially enable a broader population screening. Additionally, their affordability and remote accessibility could increase sample size and diversity in clinical trials. Nonetheless, further longitudinal studies and comparison between individuals with and without the biomarker are needed to ensure their validation in this context.

3.2.6 | Response biomarkers

Their application as response biomarkers in clinical trials and drug development requires further investigation, particularly through longitudinal studies spanning multiple years. Such studies are essential to evaluate their sensitivity to treatment or intervention-related changes over the extended timelines typical of clinical trials, an important factor for their classification within this biomarker category. However, the advantages of these tools—ability to track cognitive data at multiple time points with greater ease and lower costs—make them promising candidates under the COU of future surrogate endpoints in clinical trials. Their sensitivity to subtle cognitive changes associated with both neuropsychological tests and key AD pathological biomarkers further supports their potential utility in clinical trials. Given that primary endpoints that use standard composite cognitive measures may take longer to manifest (e.g., change from baseline in the PACC5), frequent use of digital cognitive assessment might detect earlier cognitive changes and fluctuations linked to the AD continuum progression that can be a result of drug effectiveness.

Several ongoing clinical trials are incorporating digital cognitive assessments as endpoints. For example, the ReTain clinical trial (NCT06544616) is a Phase 2b randomized, double-blind, placebo-controlled study evaluating the effects of the active immunization JNJ-64042056 in preclinical AD. This trial tests whether the treatment delays cognitive impairment by inhibiting pathological tau seeding and spreading, and it is expected to include multimodal digital biomarkers to measure cognitive and motor functions during daily activities as an exploratory endpoint. In this review the study by Papp et al. highlights the use of the C3 digital composite as a secondary endpoint to assess change in cognitive performance in the A4 trial in a preclinical AD cohort.[23] Future research will be essential to determine the sensitivity of the C3 composite to detect changes over time in the context of anti-amyloid or anti-tau drugs trials.

For digital cognitive assessments to be validated as surrogate or exploratory endpoints, it is essential that they undergo validation in clinical trial settings, demonstrating that performance and cognitive changes on these tests exhibit direct and statistically significant associations with the clinical outcomes, particularly over extended time periods.

3.2.7 | Safety biomarkers

Although digital tools that collect cognitive data provide valuable insights into cognitive function and progression, they do not currently fit within the safety biomarkers category. Safety biomarkers are primarily used to monitor adverse effects, toxicity, or safety concerns related to treatments or interventions. Therefore, they have not been validated under safety biomarkers. However, other digital tools that not only collect cognitive data can serve as safety biomarkers by collecting patient-reported outcomes (PROs) through digital platforms where patients can report side effects, symptoms, or changes in their health status as well as the use of wearable devices, which are not discussed in this review, but which enable continuous monitoring of vital signs, physical activity, and other indicators.[46,47]

4 | RECOMMENDATIONS

To select the most suitable digital cognitive assessment for a study, we recommend considering various factors that may influence its optimal use and the resulting cognitive outcomes, while also combining it with prior biomarker classification to ensure its application is tailored to the most appropriate context.

4.1 | Digital devices

Selecting the right digital device for cognitive tasks is crucial, ensuring it aligns with the task's objectives and accessibility needs.[48] Ideally, tasks should be compatible across multiple devices to maximize accessibility and avoid restricting participation to a single platform. However, device-related variability in test performance must be considered, as differences in the device could influence results.[49] If a specific device is required for the study, then it should be available to all participants, with researchers or clinicians providing it if necessary.

For frequent or continuous assessments, such as learning curve studies (e.g., BRANCH Multi-Day, ARC) or monitoring, smartphones offer convenience and ease of use. In contrast, tasks demanding higher attention, precision, or complex interactions—such as writing, mouse movements, or longer cognitive tasks—are better suited for computers or tablets due to their larger interfaces and reduced error risk. Ensuring participants are familiar with the selected device is also essential, as a lack of knowledge and experience could affect cognitive performance and introduce confounding variables to the study. Therefore, using participants' own devices might be recommended in many cases. However, the trade-off between familiarity and the variability related to performance differences between operating systems and hardware across devices should be carefully assessed.[48]

4.2 | Administration time

Administration time is another key consideration. Many digital cognitive assessments are designed to be brief, typically under 20 minutes, making them faster and easier to administer than traditional assessments.[48] Some, like automated speech biomarkers take as little as 2 to 3 minutes.[27,41] However, tasks involving delayed assessments, such as delayed memory tasks, may extend the total duration depending on the delay length. For example, the neotiv app[34] incorporates a delay in its assessments, which impacts overall administration time and should be clearly advised beforehand to the participant to not alter the cognitive results.

4.3 | Remote versus in clinic

Another important factor is whether the digital cognitive task can be administered remotely and unsupervised, or whether it requires in-clinic administration. In-clinic assessments typically require the participant's presence and clinician supervision, but some digital tools are being developed to allow for independent completion within the clinical setting. This can enable clinicians to focus on other tasks during administration, improving workflow efficiency and scalability. However, in such cases, it is essential to ensure a calm, distraction-free environment to maintain the integrity of the cognitive assessment by minimizing external stimuli and potential interruptions for the patient.[50]

Both remote and unsupervised assessments offer the advantage of eliminating the need for clinical supervision, allowing participants to complete tasks in their home environments without going to a clinic. This approach is particularly beneficial for monitoring or assessing responses to treatment, when cognitive data need to be collected over short-term periods. Additionally, they are well suited for broader population screening, as they enable access to individuals in remote or rural areas who may lack proximity to a clinic, and it reduces the need for specialized personnel to perform the cognitive testing.[11] However, remote assessments require ensuring participants are comfortable with the technology used and receive only appropriate assistance, as both factors can introduce errors and bias in the performance itself. In contrast, in-clinic settings offer greater control, minimizing these risks and ensuring a higher standardized testing environment.

4.4 | Type of cognitive test

It is as well important to consider the type of cognitive domain being evaluated. Just like traditional paper-and-pencil tests, digital cognitive assessments can be designed for specific tasks targeting particular cognitive domains or broader batteries assessing multiple domains. The choice of assessment should align with the study's objectives, with comprehensive batteries often being more suitable for in-depth evaluations.

4.5 | Language

As with traditional paper-and-pencil tests, translations of the digital task into different languages are essential to consider standardized and validated remote and digital cognitive assessments for different countries and populations. Regarding the studies included in this review, all digital tools are available to be administered in English. However, it is important to distinguish between two types of language adaptation: (1) when language is the cognitive domain being assessed, requiring a more complex process, including both linguistic and cultural adaptations to ensure cognitive equivalence, and (2) when only task instructions need translation, which is typically more feasible and straightforward, and therefore presents a lower implementation barrier. Nonetheless, both types of adaptation require time and personnel to be correctly implemented. On the other hand, digital biomarkers that may not rely on language (such as passive digital biomarkers and wearables, which are not discussed in this review) can be helpful to overcome language barriers that can sometimes arise from cognitive assessments in both research and clinical contexts.[51]

4.6 | Demographic, socioeconomic factors, and digital literacy

Other factors, such as demographic and socioeconomic characteristics, also need to be considered when performing digital cognitive assessments on the target population. While technology use is expanding, age remains a key factor, as younger generations tend to be more familiar and comfortable using digital tools.[52,53] Additionally, variables such as sex, education levels, and socioeconomic factors like income, social status, social environment or cultural context can significantly influence the effectiveness and accessibility of these assessments, being a possible limitation in countries with lower incomes and limited resources to access technology.[11,52,54,55] Closely linked to these factors is digital literacy, defined as an individual's knowledge of and familiarity with digital technology.[56] Despite its importance, few studies have incorporated a digital literacy questionnaire, and no standardized, validated instrument currently exists for use across languages, countries, and cohorts. Addressing this gap requires the administration of a brief digital literacy assessment prior to digital cognitive testing, as well as the development of a standardized questionnaire. This would ensure that participants have adequate internet access and sufficient comfort, knowledge, and familiarity with the technology required, to not alter the validity and reliability of the cognitive results, particularly in remote areas and unsupervised settings.

4.7 | Population

Finally, we also need to consider the type of population where these assessments are going to be used depending on the study. Considering previous factors (such as digital literacy or demographic and socioeconomic factors), it is expected that the older population will have less experience and familiarity with technology use. Consequently, populations with a higher proportion of individuals with dementia, advanced MCI, or co-pathologies that can significantly impact technology use may not be the most suitable for complex digital assessments. In such cases, simpler tasks—such as automatic speech assessments—may be more appropriate. While digital cognitive assessments serve as a valuable tool across various contexts, the health-care setting also plays a key role. In primary care, where efficiency is nowadays crucial, using more straightforward and fast cognitive assessments can be particularly beneficial for initial screening, helping to streamline the process. On the other hand, in secondary care, more comprehensive assessments may be required. Here, digital cognitive batteries and continuous monitoring tools are well suited, offering more in-depth cognitive profiling while optimizing time and reducing the need for specialized personnel.

5 | DISCUSSION

As digital cognitive assessments continue to evolve, emerging evidence has highlighted their utility in both AD research and clinical applications.[9,11] In this review, most studies reported moderate to high correlations between digital cognitive assessments and traditional paper-and-pencil assessments,[17–19,22–24,32–34,36,39,40,42,43] providing strong support for their concurrent validity in measuring cognitive performance. Additionally, they have shown significant associations with key AD biomarkers, such as Aβ and tau, in patterns that align with established disease pathology, and comparable to associations observed with traditional assessments. Notably, these associations have been evident even in the early stages of AD, particularly in preclinical populations including CN individuals, where early and accurate identification is crucial for both current and future clinical trials.

Within the FDA BEST framework, most digital cognitive assessments included can be classified mainly as risk biomarkers, as they are effectively associated with AD pathology in apparent CN individuals, suggesting that these individuals are at risk for future cognitive decline.[2] They also show promise as monitoring biomarkers, offering significant advantages over traditional assessments by enabling frequent, reliable, and cost-effective longitudinal cognitive evaluations. Regarding diagnostic approaches, while they are not sufficient as diagnostic biomarkers alone, they do represent a possible complementary approach for integration into AD diagnostic workflows and participant screening processes, supporting clinical routine uses and screening in clinical trials. A limited number of these digital assessments have also demonstrated utility as prognostic biomarkers, providing insights into disease progression, and some show promise for future classification as response biomarkers (endpoints) as well as predictive biomarkers, with a potential use in clinical trials. To fully establish their reliability and clinical applicability, future research should prioritize longitudinal studies with extended follow-up periods to validate their effectiveness, examine their association with disease pathology over time, and determine their role across different stages of the AD continuum.

Another key advancement in the field of AD is the growing role of blood-based biomarkers, which have become an essential part of the evolving diagnostic framework due to their non-invasiveness, cost effectiveness, and increasing diagnostic accuracy. When combined with digital and remote cognitive assessments, these biomarkers offer a promising breakthrough, with several studies included in this review already exploring the combination. This combination has the potential to provide scalable, accessible, and non-invasive solutions for AD screening, participant recruitment, diagnostic workflows, or longitudinal monitoring.[34,37] A notable real-world example advancing this approach is the AD-RIDDLE project, which is underway across sites in several countries.[57]

Despite these advancements, several limitations are also present within these assessments. A primary limitation of digital cognitive assessments is the possible variability in the data obtained, influenced by factors such as device type, internet connectivity, environmental conditions, and the participant's familiarity with technology (digital literacy), which can all impact data acquisition, accuracy, and cognitive performance itself. Additionally, socioeconomic and demographic disparities must be considered as well, as access to digital and remote assessments may be limited by these differences as well as the technological resources available. While digital cognitive biomarkers have the potential to be deployed in underserved or rural areas, it is crucial to assess whether individuals in these communities have access to these tools and possess the necessary digital literacy to complete them reliably.

Additionally, AI and machine learning algorithms, which are sometimes used in these types of assessments, are often trained on homogeneous datasets.[58] To ensure these tools yield reliable results across broader populations, validation in more diverse groups is necessary, including varied ages, races, ethnicities, and socioeconomic backgrounds, as well as sex and gender aspects. Privacy and security concerns also arise, as digital cognitive assessments often involve continuous monitoring and the storage of personal information, which needs to be well established a priori.[59]

6 | CONCLUSION

In summary, this scoping review provides an overview of current digital cognitive assessments for AD and explores their potential to enhance AD research and clinical care by enabling early detection of at-risk individuals, disease monitoring, diagnosis, prognosis, predictive insights, and response biomarker applications, while also offering a scalable and accessible approach. However, they are not exempt from limitations, which must be carefully considered to ensure their effectiveness and broad applicability.

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CONFLICT OF INTEREST STATEMENT

O.G.R. receives research funding from F. Hoffmann-La Roche Ltd and has given lectures in symposia sponsored by Roche Diagnostics, S.L.U, and Idorsia Pharmaceuticals Ltd. G.S.-B. worked as a consultant for Roche Farma, S.A. J.D.G. receives research funding from Roche Diagnostics and GE Healthcare, has given lectures at symposia sponsored by Biogen and Philips, and is currently an employee of AstraZeneca. G.S. has received speaker fees from Springer, GE Healthcare, Biogen, Esteve, and Adium. C.P.M. has nothing to disclose.The authors have no conflicts to report. Author disclosures are available in the supporting information.

CONSENT STATEMENT

Not applicable.

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